Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/20/2016
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Harderian Gland Tumorigenesis: Low-Dose-, Low-Dose-Rate-, and LET-Response		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2016, per S. Monk/LaRC (Ed., 4/5/16) NOTE: Extended to 12/31/2015, per PI (Ed., 8/13/15) NOTE: Extended to 6/30/2015 per PI (Ed., 10/8/14) NOTE: Gap change to delete Cancer 05, per PI (Ed., 12/17/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chang, Polly (SRI International)		
Grant/Contract No.:	NNJ11HA94I		
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Task Description:	Increased cancer risk remains a primary concern for travel into deep space and may preclude manned missions to Mars due to large uncertainties that currently exist in estimating cancer risk from the spectrum of radiations found in space with available human epidemiological radiation-induced cancer data. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluences of radiations found in space using radiation quality factors and dose-rate modification factors, and assuming linearity of response since the shapes of the dose-responses at low doses below 100 mSv are unknown. The goal of this proposal is to reduce uncertainties in the relative biological effectiveness/linear energy transfer (RBE/LET) relationship for particle radiation carcinogenesis. Our approach is to use the existing data on Harderian gland tumorigenesis as reference, and obtain Harderian gland (HG) tumor data for heavy ions beams not previously reported to fill the gaps in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our understanding of the dose-response curve at low doses and to test for deviations from linearity. The hypothesis to be tested is that filling in these data gaps will delineate more comprehensively the space radiation risk of space radiation induced tumorigenesis, in particular to provide data that will allow us to test for a role of targeted versus non-targeted effects in cancer risk. We aim to investigate the effects of a broader range of radiation qualities, doses, and dose rates not previously reported on the development of Harderian gland and also lung tumors in ~100-120 day old female CB6F1 mice 16 months after exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of the proposed work will fill a significant gap in Harderian tumor response data at low fluences of Silicon or Titanium particle beams to fill in the tumor prevalence in the LET range from below 25 keV/micrometer and up to approximately 193 keV/micrometer to improve our understanding of the dose-response curve at low doses relevant to space travel and to test for deviations from linearity. The data should contribute to decreasing the current uncertainty in estimating radiation risk of cancer due to exposures in space travel.
Task Progress:	The analysis of our results reveals that the sham-treated, but unirradiated, animals have a low HG tumor incidence of 4.97±2.31%, which is comparable to the previously published control incidence of Alpen et al. 1993 of 2.6±2.5%, and confirms our replication of the same animal strain and protocols. In addition, both the gamma ray, and the iron ion studies we completed were nearly identical to those reported by Alpen et al. The dose-dependent increase in HG tumor incidence after single low doses of 260 MeV/u Silicon ions at 70 keV/µm or 1 GeV/u Titanium ions at LET of 100 keV/µm matched the trends of the historical data from Alpen et al., and nested between the Alpen 600 Mev/u Iron and 670 MeV/u Neon data. The results were used to test two models. The targeted (TE) model assumes a linear dose response at low doses and the non-targeted (NTE) model that assumes a nonlinear threshold type response in addition to the linear dose term at low doses. Unlike many of the animal model systems used for high-LET tumorigenesis studies, this established model system has a low and consistent spontaneous background for HG tumors, allowing us to measure tumor frequency increases at particle doses below 0.1 Gy with a reasonable number of animals in each dose group. It is this low-dose region that is most important for space radiation protection considerations. Our results showing that nontargeted effects provide an improved fit to these data, and the large deviation between TE and NTE models in this important low-dose region, suggest that possible mechanisms for NTE on tumor induction should be an important focus of future studies. A detailed description of the results from the particle-radiation induced Harderian Gland tumorigenesis studies can be found in our recently published paper in the journal Radiation Research (Radiat Res. 2016 May;185(5):449-60). Overall, the composite space radiation-induced HG tumorigenesis data and theoretical modeling reported in our publication represent a comprehensive documentation of evidence supporting a role for nontargeted effects for particle exposures involving one or less track per cell nucleus. Since it is not feasible to carry out very low-dose-rate protracted exposures at the NSRL (NASA Space Radiation Research Laboratory) at this time, we exposed animals to 5 dose fractions of very low doses (0.026 or 0.052 Gy) of 1,000 MeV/u titanium ions delivered in five successive days, and compared the tumor prevalence with animals that received a single acute dose of 0.13 or 0.26 Gy titanium ions. We found no significant difference in tumor incidence between the two dosing regimes. The interval between irradiations has been reported to be important in carcinogenesis; however, those studies were generally done using higher dose (>0.2 Gy), and in some cases with dose fractions separated by 30 days. Our results suggest that at space-relevant doses, it is likely that dose-rate effects do not occur. However, further studies using either very low-fluence strategies or doses over longer intervals between exposures may be required to completely rule out the impact of low dose rates in protracted exposures. Since animals were whole-body irradiated, we were also able to evaluate the lung tumor incidence and histology from these animals. Pooled spontaneous lung tumor frequency from sham-treated animals was 9.24±2.27%. The dose response of the Si-induced lung tumors revealed non-linearity with an early steep dose response, followed by a nadir at 16 cGy, and then an increase at 32 cGy. Results from the Titanium study suggested that there was no statistically significant increase in lung tumor incidence with increasing dose compared to the unirradiated controls. Animals irradiated with Iron ions showed a dramatic increase in tumor frequency up to 24.3±9.7% after 10 cGy, a decline to 10.6±9% after 20 cGy, and elevation up to 26±13.8% after 40 cGy. As noted, although >40 animals were used for each dose group, the error bars are large and the significance of these findings are unclear at this point. A variety of non-HG non-lung tumors were discovered during terminal necropsy of these animals. These include mammary carcinomas, granulocytic myelocytomas in the liver sinusoids, carcinomas, histiocytic sarcomas in the spleen, undifferentiated sarcomas near the salivary glands, postorbital lymphosarcoma, etc. Some are primary tumors while other appeared to be due to metastatic disease. The frequency of all ‘other tumors’ appear to increase with increase in dose for the Si, Ti, Fe, and Cs beams. Colon tissues were collected from the Silicon-irradiated animals per a protocol provided to us by Dr. Jerry Shay’s group at University of Texas SW (UTSW) in Dallas, TX. These tissues were shipped to Dr. Shay’s laboratory to be evaluated. Results show that there were no polyps or colon tumors in any of the low Silicon ion irradiated animals 16 months after radiation exposure. Reference Alpen EL, Powers-Risius P, Curtis SB, et al. Tumorigenic potential of high-Z, high-LET charged-particle radiations. Radiat Res 136, 382-391. 1993.

The following presentations were made during 2015:

Chang PY, Bjornstad KA, Bakke J, Rosen C, Du N, Fairchild D, and Blakely, EA, Update of Harderian Gland Tumorigenesis: Low - Dose and LET - Response. NASA Human Research Program (HRP) Investigator's Workshop, Jan 11 - 15, 2015, Galveston, Texas — poster presentation.

Blakely, EA, "Validation of Biological Data from Experiments with Individual Ion Beams to Predict Risk from Complex GCR Fields, 2015 Human Research Program and NASA Investigators' Workshop, Galveston, TX, Jan 14, 2015.

Blakely EA, "Review of Biology Related to Heavy - Ion Radiotherapy", HIMAC International Symposium, Fujisoft Akiba Plaza, Tokyo, Japan, Jan 19 - 20, 2015. Jan 19, 2015.

Blakely EA, "Charged Particle Biology & Radiation Therapy", Jiangsu Proton and Heavy Ion Hospital, Nanjing City, Jiangsu Province, January 22, 2015.

Blakely EA, "Radiobiology of Proton and Heavy - Ion Therapy", UCSF, Radiation Oncology, San Francisco, CA, January 29, 2015.

Blakely EA, "Radiobiology of Protons and Heavy - Ion Therapy", 2015 Translational Cancer Research Seminar Series, Cincinnati Children's Hospital Medical Center, Cincinnati, OH, March 12, 2015.

Dauer LT and Blakely EA , "Guidance on Radiation Dose Limits for the Lens of the Eye" Status of NCRP SC 1 - 23 Commentary, 2015 51st Annual Meeting of the NCRP, Changing Regulations and Radiation Guidance: What does the future hold? March 16 - 17, 2015, Bethesda, MD.

Blakely EA and Chang PY, "Particle Radiation - Induced Harderian Gland Tumorigenesis", Space Radiation and Heavy Ion Therapy Symposium, (SRHITS2015), Abeno Harukas, Osaka, Japan, May 22 - 24, 2015.

Chang PY, Bjornstad KA, Bakke J, Rosen C, Du N, Fairchild D, and Blakely, EA , Harderian Gland Tumorigenesis: Low - Dose - and LET - Response" , 13th International Congress for Radiation Research, Kyoto Japan May 24 - 30, 2015 --- poster presentation.

Blakely EA, "Harderian Gland Tumorigenesis: Low - Dose - and LET - Response", 13th International Congress of Radiation Research, Kyoto, Japan, Early & Late Biological Effects of Space Radiation Symposium, May 28, 2015.

Blakely EA, "Radiation and Cataracts", UCSF, Radiation Oncology Residents, San Francisco, CA, June 4, 2015.

Blakely EA "Cataracts" 2015 NASA Space Radiation Summer School, Brookhaven National Laboratory, Upton, NY, June 25, 2015.

Blakely, EA, "Charged Particle Biology & Radiation Therapy", NASA Space Radiation Summer School, Brookhaven National Laboratory, Upton, NY, June 25, 2015.

Blakely, EA, "Historical Review of Accelerator - Based Radiation Therapy, ERHS751 - Advanced Radiation Biology, MRB#463 - Videocast, December 10, 2015.

Bibliography Type:	Description: (Last Updated: 02/12/2026)
Articles in Peer-reviewed Journals	Chang PY, Cucinotta FA, Bjornstad KA, Bakke J, Rosen CJ, Du N, Fairchild DG, Cacao E, Blakely EA. "Harderian gland tumorigenesis: Low-dose and LET response." Radiat Res 2016 May:185(5):449-60. http://dx.doi.org/10.1667/RR14335.1 ; PubMed PMID: 27092765 , May-2016
Articles in Peer-reviewed Journals	Barcellos-Hoff MH, Blakely EA, Burma S, Fornace AJ Jr, Gerson S, Hlatky L, Kirsch DG, Luderer U, Shay J, Wang Y, Weil MM. "Concepts and challenges in cancer risk prediction for the space radiation environment." Life Sci Space Res. 2015 Jul;6:92-103. http://dx.doi.org/10.1016/j.lssr.2015.07.006 , Jul-2015
Articles in Peer-reviewed Journals	Garikipati VNS, Arakelyan A, Blakely EA, Chang PY, Truongcao MM, Cimini M, Malaredy V, Bajpai A, Addya S, Bisserier M, Brojakowska A, Eskandari A, Khlgatian MK, Hadri L, Fish K, Kishore R, Goukassian DA. "Long-term effects of very low dose particle radiation on gene expression in the heart: Degenerative disease risks." Cells. 2021 Feb 13;10(2):387. https://doi.org/10.3390/cells10020387 ; PMID: 33668521; PMCID: PMC7917872 , Feb-2021
Articles in Peer-reviewed Journals	Leon Parada KN, Lawson G, Chang P, Blakely EA, Bajwa L, Gramajo-Aponte K, Getze S, Luderer U. "Irradiation with a mixed heavy ion beam induces ovarian follicle loss and dose-dependent mixed ovarian tumor development." Biol Reprod. 2026 Jan 21:ioag019. https://doi.org/10.1093/biolre/ioag019 ; PMID: 41562572 , Jan-2026
Awards	Blakely E. "2015 Berkeley Laboratory Director's Award for Exceptional Achievement, November 2015." Nov-2015